



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Decision-Making and Resource Optimization with Linear Programming

TGS Ref No: TGS-2023036656

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 6.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
5.0	10 Jan 2024	Legacy learner-guide slide deck (153 slides, rasterised tables).	Dr Alfred Ang
6.0	28 Aug 2026	Rebuilt in the current visual, mechanism-led house style: 147-slide deck, 8 progressive Excel labs, single-source content. Lesson Plan re-aligned to the exact deck slide numbers.	Dr Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Assessment.....	4
Course Schedule.....	4
Day 1 — Linear Programming, the Excel Solver and Decision Analysis.....	4
Day 2 — Risk, Multi-Criteria Decision Analysis, AHP and Assessment.....	5
Lesson-to-Slide & Lab Alignment.....	6

Course Information

Course Title	Decision-Making and Resource Optimization with Linear Programming
WSQ Course Reference	TGS-2023036656
Skills Framework (TSC)	Resource Management — WST-SPI-4006-1.1-1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Microsoft Excel labs (LP / Excel Solver / decision analysis / AHP)
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1 — Develop a resource-management strategy and allocation plan using Linear Programming (LP) and the Excel Solver.
- LO2 — Develop a resource-allocation plan and track resource usage using Decision Analysis (Expected Value and Expected Utility).
- LO3 — Evaluate resource-management performance and set KPIs using Multi-Criteria Decision Analysis (scoring board and the Analytic Hierarchy Process).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book (tests K1-K3).
- Practical Performance (PP) — hands-on Excel tasks, 1 hour, open book (tests A1-A9).
- Format: Open Book — course slides, Learner Guide, lab workbooks and approved materials only.
- The final assessment is conducted on Day 2 from 3:45 pm (briefing), with the WA and PP from 4:00–6:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

The 'Deck slides' column references the exact slide numbers in the current trainer deck (Decision-Making and Resource Optimization with Linear Programming-v6.0.pptx, 147 slides), keeping the Lesson Plan and the deck aligned.

Day 1 — Linear Programming, the Excel Solver and Decision Analysis

Time	Duration	Topic / Activity	Deck slides
9:00–9:30	30 min	Welcome, course introduction, ground	2–8

		rules, downloading course material, and mandatory digital attendance (AM)	
9:30–9:45	15 min	Skills framework (Resource Management), knowledge & abilities, learning outcomes, course outline and lesson plan	9–17
9:45–10:45	60 min	Topic 1 — Optimisation & Linear Programming: the LP model, standard form, the five assumptions and the Lucy's Cupcakes formulation	18–33
10:45–11:00	15 min	Tea break	—
11:00–12:30	90 min	Hands-on: Lab 1 (formulate an LP model) and Lab 2 (solve for maximum profit with the Excel Solver, Simplex LP)	34–46
12:30–13:30	60 min	Lunch break	—
13:30–14:15	45 min	Reading the Answer Report, binding vs non-binding constraints, slack, shadow prices and sensitivity analysis	47–51
14:15–15:15	60 min	Hands-on: Lab 3 (sensitivity & scenario analysis); LP failure diagnostics (infeasible / unbounded / degenerate)	52–62
15:15–15:30	15 min	Tea break	—
15:30–17:30	120 min	Topic 2 — Decision Analysis: decision trees, Expected Value; Hands-on: Lab 4 (EV, EVwPI, EVPI)	63–78
17:30–18:00	30 min	Day 1 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours), excluding the 1-hour lunch.

Day 2 — Risk, Multi-Criteria Decision Analysis, AHP and Assessment

Time	Duration	Topic / Activity	Deck slides
9:00–9:15	15 min	Day 1 recap and mandatory digital attendance (AM)	—
9:15–10:15	60 min	Topic 2 — risk &	79–90

		Expected Utility; Hands-on: Lab 5 (risk and expected utility, certainty equivalent)	
10:15–10:30	15 min	Tea break	—
10:30–12:30	120 min	Topic 3 — MCDA & the scoring board (Lab 6); the AHP hierarchy, pairwise comparison and criteria priorities	92–102, 103–111
12:30–13:30	60 min	Lunch break	—
13:30–15:00	90 min	Hands-on: Lab 7 (AHP consistency, CR) and Lab 8 (integrated resource-allocation evaluation & KPIs)	112–129
15:00–15:45	45 min	Method selection (LP vs Decision Analysis vs MCDA), course recap, Q&A and PM digital attendance	130–143
15:45–16:00	15 min	Briefing for Assessment	16, 144
16:00–17:00	60 min	Written Assessment (WA) — Short- Answer Questions (SAQ), 1 hour, open book (tests K1–K3)	—
17:00–18:00	60 min	Practical Performance (PP) — hands-on Excel tasks across LP, DA and MCDA, 1 hour, open book (tests A1–A9); TRAQOM survey and assessment digital attendance on completion	144–147

Total training time: 480 minutes (8 hours), excluding the 1-hour lunch.

Lesson-to-Slide & Lab Alignment

Each topic and lab below cites its exact slide range in the trainer deck and its hands-on workbook, so any change to the deck can be re-checked against this table.

Topic / Lab	Maps to (K/A)	Deck slides	Lab workbook
Topic 01: Optimisation and Linear Programming for Resource Management	K1, A1, A2, A3	18–62	
Lab 1: Formulate a Linear Programming Model	A1, A3, K1	34–37	lucys_cupcakes.xlsx
Lab 2: Solve for	A2, K1	41–46	solver_maxprofit.xlsx

Maximum Profit with Excel Solver			
Lab 3: Sensitivity and Scenario Analysis	A2, A3, A6	52–56	sensitivity_scenario.xlsx
Topic 02: Decision Analysis for Resource Usage	K2, A4, A5, A6	63–91	
Lab 4: Expected Value, EVwPI and EVPI	A4, A5, K2	74–78	kazo_decision.xlsx
Lab 5: Risk and Expected Utility	A6, K2	86–90	risk_utility.xlsx
Topic 03: Multi-Criteria Decision Analysis	K3, A7, A8, A9	92–135	
Lab 6: Weighted Scoring Board (MCDA)	A7, K3	98–102	supplier_scoring.xlsx
Lab 7: AHP — Criteria Priorities and Consistency	A8, K3	112–118	ahp_criteria.xlsx
Lab 8: Integrated Resource-Allocation Evaluation & KPIs	A8, A9, K3	123–129	integrated_eval.xlsx